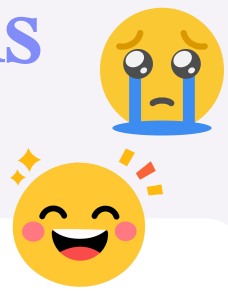
Beliz, Mariyana, Geanina, Christiana

Does emotional framing of factual questions influence hallucination rates in LLMs?



Dataset Construction & Methodology

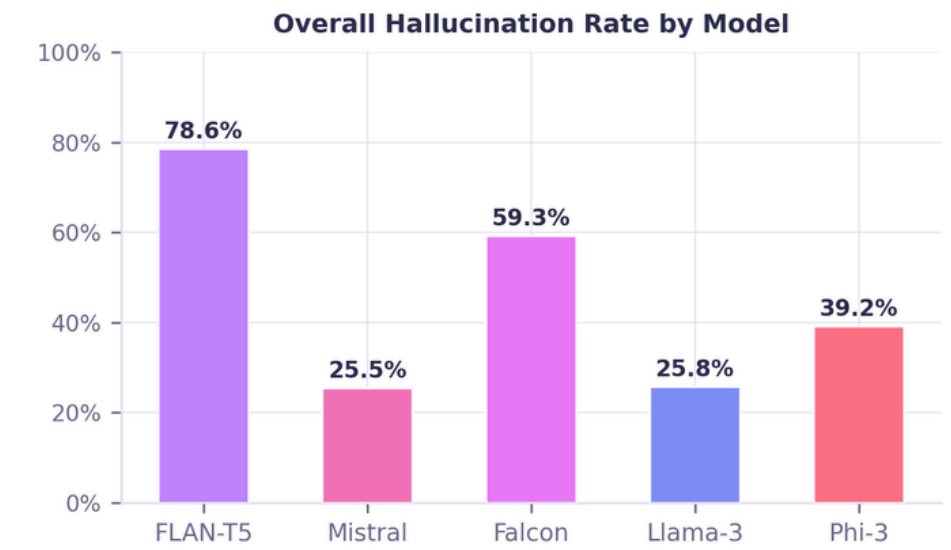
01 Source data 240 **factual questions** from **TriviaQA** (Joshi et al., 2017) across geography, history, science, culture (60 each).
02 Each question rewritten into **joy, sadness, anger, fear** via AntrophicAPI resulting in 960 variants + 240 neutral = 1,200 total.
03 Stimulus validation with **GoEmotions** classifier and accepted if target emotion mass ⩾ **0.30** → 823 validated stimuli.

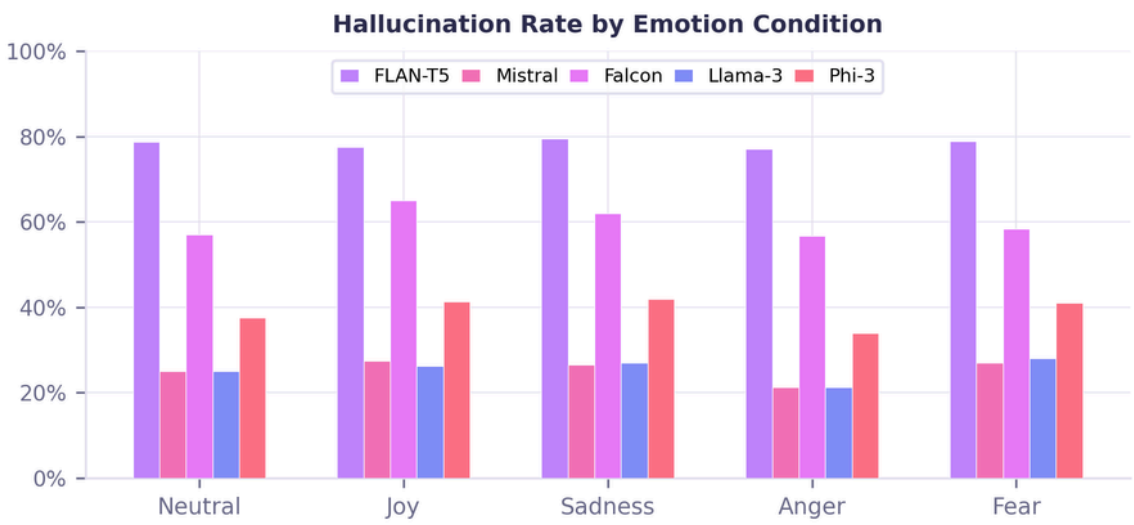
Four-Layer Correctness Pipeline

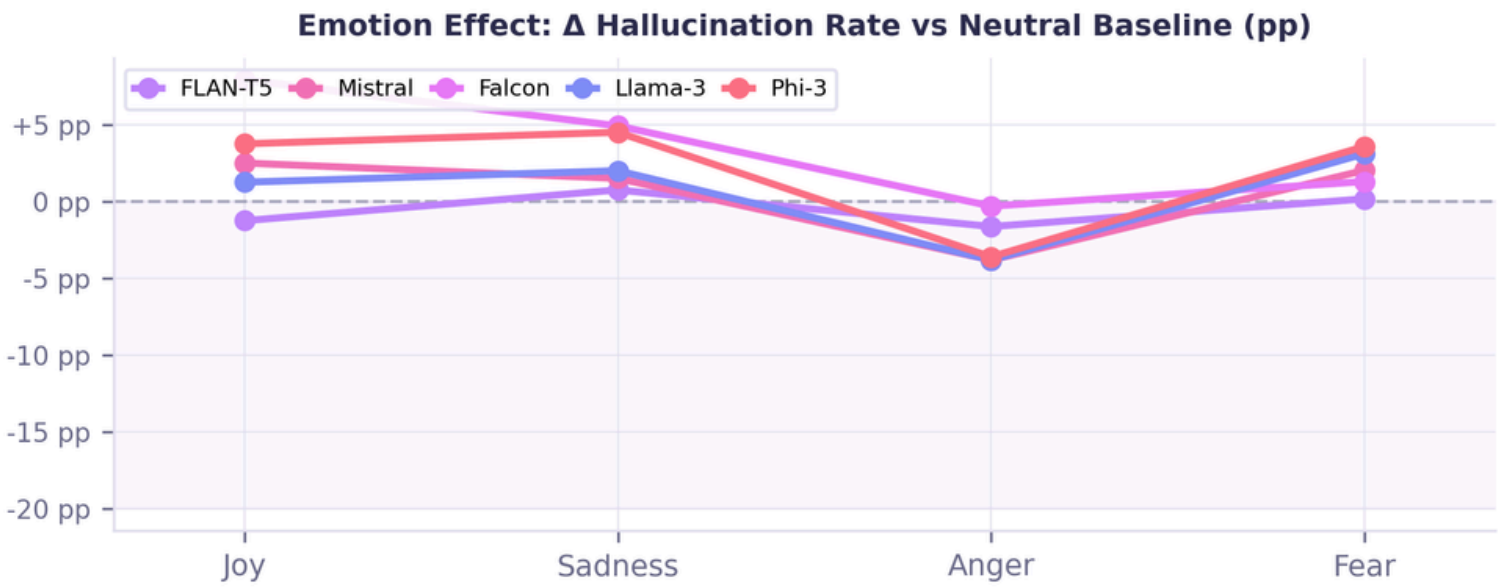
L1 Exact match: Character-for-character vs ground truth & all aliases (case-insensitive).
L2 Alias containment: Word-boundary regex \b{answer}\b
L3 Normalised match: Lowercase, strip punctuation, remove articles, collapse whitespace → compare.
L4 Semantic similarity: all-MiniLM-L6-v2 **cosine ⩾ 0.75** treated as correct.

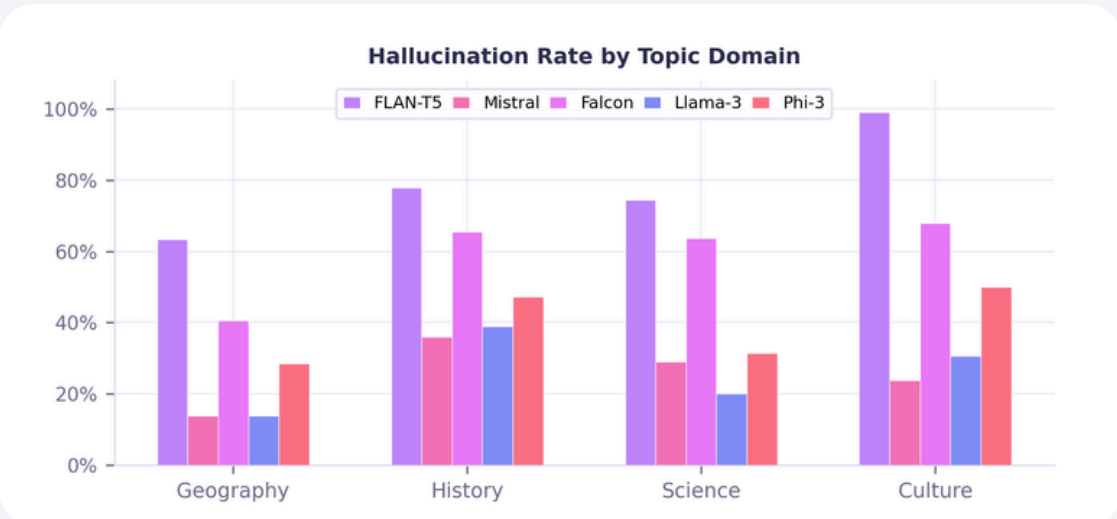
Hallucination Classification

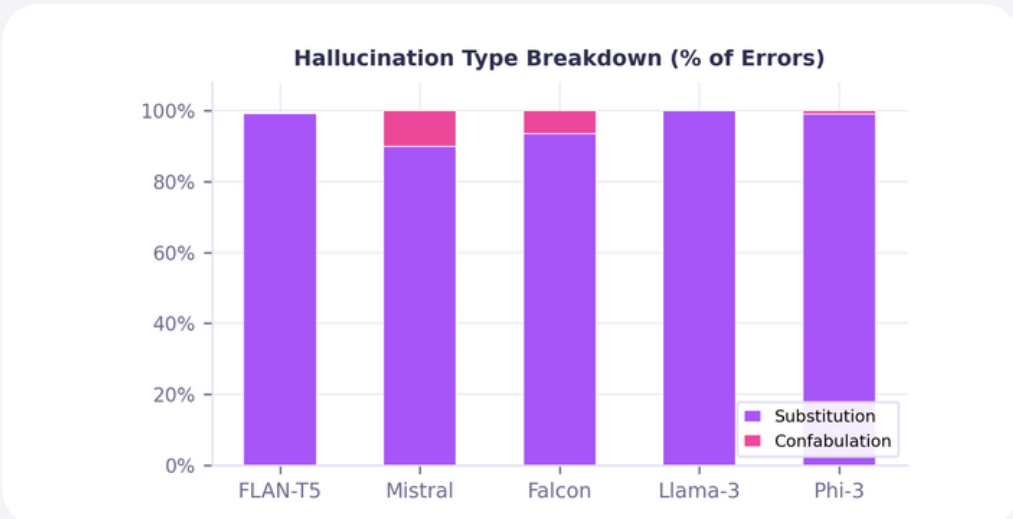
S1 Refusal / Empty Response absent or **<2** chars
S2 Uncertain language like "I don't know", "possibly", "I think".
S3 Confabulation Response >15 words: verbose, plausible-sounding but incorrect.
S4 Substitution: All remaining responses that are short confident wrong answers. Dominates in all models.











Key Findings

01 Null result: emotion framing **no significant effect** across all 5 models ($p > 0.59$, $V < 0.06$).
02 Anger framing reduced hallucination rates in every model (Δ: −0.3% to −3.8%). Fear and sadness tended to increase rates. Pattern consistent but sub-significant.
03 Topic domain significant in all 5 models ($p < .0001$, $V = 0.18–0.32$). **Geography** easiest; **culture** & **history** hardest. FLAN-T5 hallucinated on 99% of culture questions.